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# From Pair Scores to Prerequisite Graphs on Skill-Taxonomy Descriptions: Evidence Format, Lexical Priors, and Graph Constraints

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Prerequisite graphs support curriculum design and skill-based learning paths, yet language-model predictors usually stop at a pairwise score that cannot be audited as a graph. We describe a seven-step tool that turns those scores into a directed acyclic graph whose edges carry a verbatim skill-card span. On the public ESCO-PrereqSkill release, under an unseen-concept split, we separate prompt format, a lexical mention rule, temperature scaling, and cycle constraints for `gpt-4o-mini` and Qwen 2.5 7B. Presenting the skill card as evidence improves ranking for `gpt-4o-mini` (ROC-AUC 0.886) and lowers it for Qwen. A mention rule remains the strongest standalone ranker; stacking it with the model scores, fit on the training pairs, reaches PR-AUC 0.260. The same constraints then yield two graphs: a higher-recall graph from the taxonomy prompt (graph F1 0.124) and a higher-precision graph from the stack (graph F1 0.198). On taxonomy-derived gold, a lexical rule is a strong prior, and the language model is most useful as a complementary re-ranker.

*Keywords*: Prerequisite graph; large language models; skill taxonomy; verbalized confidence; calibration; directed acyclic graph.

## 1. Introduction

A prerequisite relation specifies which competence must be acquired before another can be taught or assessed. Curriculum designers, maintainers of a skill classification

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such as ESCO, and learning-path recommenders need that relation as an edge they can inspect: a direction, a span of source text, and a place in an acyclic graph. Recent pairwise predictors, including LECDPR [1] and ProPRL [2], optimize a binary label and stop at a score. AutoPRE [3] and KnowLP [4] already use large language models (LLMs) for concept prerequisites. A score still leaves reverse conflicts and cycles unresolved, so it is not yet a graph a practitioner can audit.

We study that conversion on skill-taxonomy descriptions. The aim is diagnostic: each stage is isolated so that its contribution is visible. Three questions organize the paper.

- **RQ1.** Does presenting a taxonomy description as evidence improve pairwise ranking, and does the effect depend on the model?
- **RQ2.** Can LLM scores add value beyond a lexical mention rule?
- **RQ3.** What do temperature scaling and structural constraints contribute once those scores are turned into a graph?

The contributions answer those questions at the level of the tool.

(i) A seven-step induction procedure whose output edge carries evidence, a verbalized score, a direction, and a type, together with an explicit map from each reported metric to the score field it uses.
(ii) An ablation of prompt format against a mention rule and a logistic stack, on an unseen-concept split of ESCO-PrereqSkill [5].
(iii) Two acyclic graphs from the same constraints: one induced from the taxonomy prompt, and one from the stack. The first recovers more gold edges; the second is the higher-precision output.

Presenting the skill card as evidence raises pairwise ranking for `gpt-4o-mini` and lowers it for Qwen 2.5 7B. A mention rule is the strongest standalone ranker, and a stack of that rule with the model scores, fit on the training pairs, reaches PR-AUC 0.260. The same structural constraints then yield a higher-recall taxonomy graph and a higher-precision stack graph. Section 2 places the tool among pairwise predictors, taxonomy induction, and verbalized confidence. Section 3 defines the edge. Section 4 gives the data flow, the constraints, and the metric map. Section 5 describes the release, the split, and the author-labeled sample. Section 6 answers RQ1–RQ3. Section 7 interprets the model dependence and the practical choice between the two graphs.

## 2. Related Work

### 2.1. *Pairwise prerequisites*

RefD [6] measured prerequisite signals in Wikipedia and textbooks. Later work recovered course dependencies [7], learned concept graphs from click and catalog data [8, 9], and extracted relations from MOOCs [10], online resources [11], and

course descriptions [12]. Supervised and graph-neural models add learning-path supervision [13], graph message passing [14], concept-pair networks [15], and course-map convolution [16]. LECDPR [1] retrieves text and then applies a graph neural network over concept–document interactions. ProPRL [2] adds a training loss that penalizes high scores in both directions. These systems stop at a pair decision or a learned graph over educational documents. Their published F1 figures belong to their own protocols, so this paper compares stages inside one protocol rather than ranking those figures against one another.

### 2.2. *Skill taxonomies and taxonomy induction*

ESCO is the European skill classification; text-mining audits ask whether it tracks industrial change [17]. Occupational studies recover nested skill dependencies [18], and known course prerequisites have been used to order modules toward jobs [19]. KnowEdu [20] builds educational knowledge graphs from pedagogical data. Le and Abel [5] released ESCO-PrereqSkill and compared zero-shot models with expert links by generating prerequisite skill names and aligning them with semantic similarity. A later system tags learning resources, keeps an evidence span, and applies graph constraints on a competency graph [21]. Taxonomy expansion, as in Taxo-Expan [22], asks whether a new concept is a hyponym of an existing anchor. That line of work grows a taxonomy. The present task orients prerequisite edges inside a taxonomy that is already given, using each skill's own card as the only document. LectureBank [23] is a slide corpus for NLP education; the public files available here contain topic titles, so slides remain future evidence rather than a second result table.

### 2.3. *Verbalized confidence*

Modern neural networks are often miscalibrated, and temperature scaling is a standard post-hoc correction [24]. For RLHF-tuned language models, a probability written into the answer can be better calibrated than the model's token probabilities [25]. A broader evaluation finds that verbalized confidence is frequently over-confident, and that the quality of the elicited number depends on the model and on the prompt [26]. The scorer in this paper is asked for two such numbers: a directed probability and a separate confidence. They are the model's own JSON fields, discrete in steps of 0.1, and they need not rank pairs in the same order. Section 4 assigns each metric to one of them. Retrieval-augmented generation retrieves text and then writes [27]; here the retrieved object is a verbatim card span, and model-written text is never stored as evidence.

### 2.4. *Unseen concepts*

Closed-set splits inflate apparent generalization when the same concepts appear in training and test [28]. The headline split therefore holds out skills. Textbook

annotation protocols show that agreement depends on explicit criteria [29]. The 240-pair sample in Section 5.3 is an author pilot with a written rubric; a study with external raters remains open.

## 3. Problem

Let $V$ be a set of skills. Each skill has a stable identifier, a preferred label, a description, and, when the release provides one, a taxonomy path. The corpus is the set of those cards. The output is a directed acyclic graph on $V$. An accepted edge from $s_i$ to $s_j$ stores a verbatim span, the scaled directed probability $\tilde{p}_{AB}$, an uncertainty $1 - \tilde{c}_{AB}$ derived from scaled verbalized confidence, a type in $\{\text{direct}, \text{transitive}, \text{none}, \text{ambiguous}\}$, and an optional review state. Table 1 fixes the symbols used below. The JSON field names appear only in Appendix A.

Table 1. Symbols. For a stored ordered pair $(A, B)$, $A$ is the candidate prerequisite.

| Symbol | Meaning |
|---|---|
| $\hat{p}_{AB}$, $\hat{p}_{BA}$ | Verbalized directed probabilities |
| $\hat{c}_{AB}$ | Verbalized confidence for the pair |
| $\tilde{p}_{AB}$, $\tilde{c}_{AB}$ | The same fields after temperature scaling |
| $\tau$ | Acceptance threshold on $\tilde{p}_{AB}$ |
| $\mu$ | Margin used when both directions exceed $\tau$ |
| $\delta$ | Band below which both directions are kept |

Candidate generation does not enumerate every ordered pair, so graph recall is bounded by the fraction of gold edges that enter the pool. That ceiling is reported with every graph.

## 4. The Induction Tool

### 4.1. *Data flow*

Figure 1 shows the objects that move through the tool. Skills are canonicalized with stable identifiers. An inverted index then proposes candidates from shared bigrams, token containment, and token overlap of at least two, without reading gold edges. For each ordered pair the tool retrieves up to four verbatim windows of 280 characters in each direction, preferring windows that contain order words such as *before* or *requires*. One decoding at temperature 0 returns both directed probabilities, a confidence, a type, and a direction. Temperature scaling [24] is fit on the validation split. Algorithm 1 then suppresses the weaker reverse direction and deletes edges until the graph is acyclic. A final queue ranks pairs for human review by uncertainty or by confidence. Only the prompt that includes the card and the taxonomy path continues through scaling into the published graphs. The other prompts are scoring ablations.

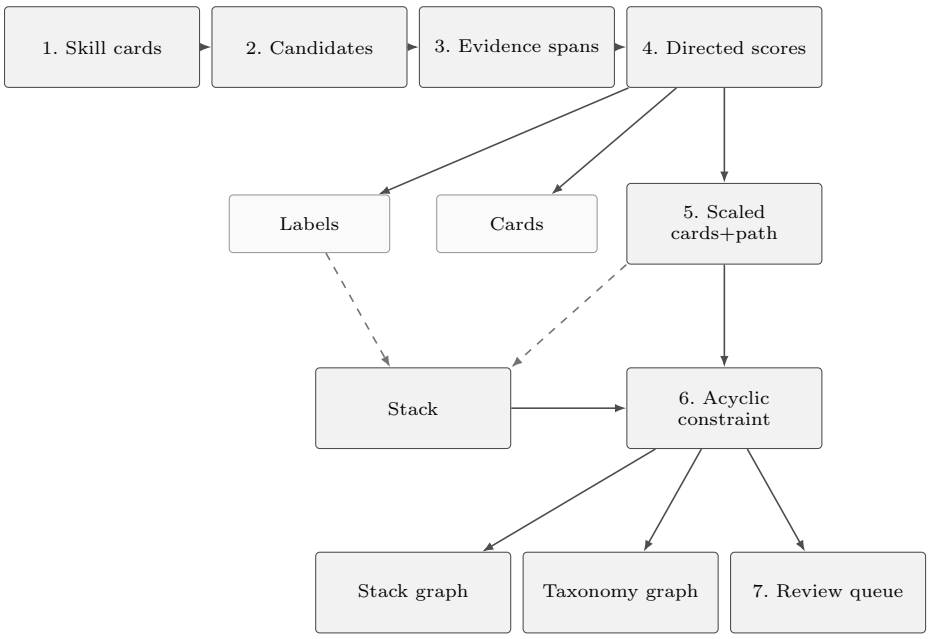



Fig. 1. Seven-step data flow, with two graph branches. Labels and Cards stop after the directed scores. Cards+path is scaled, then constrained. The stack combines the mention rule with the Labels and Cards+path scores and enters the same acyclic step. Dashed arrows mark those score inputs.

Table 2 names the comparisons. In the scoring logs the three prompts are stored as B1, B3, and B5; the reproduction note in Appendix B records that map.

Table 2. Conditions compared in the experiments. The graph row transforms scores already obtained; it does not call the model again.

| Condition | Prompt | Scaling | Graph |
|---|---|---|---|
| Labels | label and description | no | no |
| Cards | same text as a retrieved span | no | no |
| Cards+path | span plus taxonomy path | no | no |
| Scaled cards+path | span plus taxonomy path | yes | no |
| Graph | span plus taxonomy path | yes | yes |
| Stack | mention, Labels, Cards+path | logistic | optional |

### 4.2. *From scores to an acyclic graph*

Let $\tau$ be the operating threshold, let $\mu = 0.8$, and let $\delta = 0.05$. Self-loops are dropped. If both directed probabilities exceed $\tau$ and differ by less than $\delta$, the pair is marked as a direction conflict and both values are kept. Otherwise the weaker

**Algorithm 1** Reverse-conflict suppression and cycle breaking.

**Require:** scored pairs; threshold $\tau$; margin $\mu$; band $\delta$

**for** each ordered pair $(A, B)$ **do**
  **if** $A = B$ **then**
    drop the pair
  **else if** $\tilde{p}_{AB} > \tau$, $\tilde{p}_{BA} > \tau$, and $|\tilde{p}_{AB} - \tilde{p}_{BA}| < \delta$ **then**
    mark a direction conflict and keep both values
  **else**
    replace the weaker value by $\min(\tilde{p}_{\text{weak}}, \max(0, \mu - \tilde{p}_{\text{strong}}))$
  **end if**
  **if** $\tilde{p}_{AB} \geq \tau$ **then**
    accept $A \to B$
  **end if**
**end for**
**while** a strongly connected component has more than one vertex **do**
  delete its lowest-$\tilde{c}$ internal edge
**end while**
**return** the remaining edges

value is replaced by $\min(\tilde{p}_{\text{weak}}, \max(0, \mu - \tilde{p}_{\text{strong}}))$. Edges with $\tilde{p}_{AB} \geq \tau$ enter the graph. While a strongly connected component has more than one vertex, the edge of lowest scaled confidence is deleted; ties break by source identifier, then by target identifier. A taxonomy flag records a high-confidence cross-branch pair ($\tilde{c}_{AB} > 0.8$) and leaves the edge in place. The type label that would separate a direct edge from a transitive shortcut is defined so the schema is complete, and it is left unused in the tables: annotators almost never chose it (Section 5.3). The quantity $L_{\text{irr}} = \text{mean}\max(0, \tilde{p}_{AB} + \tilde{p}_{BA} - \mu)$ on gold-positive pairs follows the irreversibility penalty of ProPRL [2] as a post-hoc summary, with no gradient step. Algorithm 1 states the procedure used for every reported graph. Batch sizes and deletion counts are in Appendix B.

### 4.3. *Which field each metric uses*

Verbalized confidence and the directed probability answer different questions. Confidence is a single number for the pair; the directed probability is what the graph thresholds. Table 3 is the map used throughout the results. Pairwise ranking tables use verbalized confidence. That field is the model's own certainty about the pair, and both calibration and the pair-level threshold are defined on it. The directed probability answers a different question, the chance of one direction, and it is the field the graph thresholds. On the taxonomy prompt the directed probability does rank the test pairs better (PR-AUC 0.120 against 0.101 for confidence; Appendix C). The two fields still order those pairs in the same direction. Cycle deletion follows scaled confidence, because that is the uncertainty attached to the edge. The

acceptance decision follows the directed probability.

Table 3. Metric-to-field map. Temperature scaling is monotone, so it leaves ROC-AUC and PR-AUC unchanged and moves only thresholded F1.

| Metric | Field |
|---|---|
| ROC-AUC, PR-AUC, F1 at 0.5 | $\hat{c}$ |
| F1 at $\tau$ for Cards and Cards+path | $\tilde{c}$ |
| Direction accuracy | $\hat{p}_{AB}$ against $\hat{p}_{BA}$ |
| Edge acceptance | $\tilde{p}_{AB}$ |
| Cycle-deletion order | $\tilde{c}$ |
| ECE, Brier | $\hat{c}$ and $\tilde{c}$ |
| Graph F1 | $2\,\mathrm{TP}/(\lvert E\rvert + \lvert\mathrm{gold}\rvert)$ |

Pair ROC-AUC, PR-AUC, and F1 at $\tau$ on the test split are reported with 95% bootstrap intervals from 1,000 resamples (seed 42). With a 2% positive rate, PR-AUC is the ranking summary we emphasize. Direction accuracy is computed only on gold edges oriented $A \to B$.

## 5. Experimental Setup

### 5.1. *Data, split, and models*

The corpus is the public ESCO-PrereqSkill release [5]. After conversion there are 5,628 skills, 5,817 gold-edge records (5,797 unique directed pairs), and 5,627 cards. One unresolved name has an identifier and no card. The primary candidate pool contains 135,726 directed pairs and covers 2,800 of the 5,797 gold edges (48.3%). Split E follows Zhu and Zamani [28]: any pair that touches a held-out skill is reserved for test. The training slice has 77,507 pairs, validation has 8,611 pairs, and the test slice has 49,608 pairs with 1,003 positives. A closed split that reuses concepts is used only as a robustness check in Appendix C. `gpt-4o-mini` is the model used for every published graph. Qwen 2.5 7B Instruct is scored on the same split as a second family. Both decoders use temperature 0. The prompts are in Appendix A.

### 5.2. *Lexical baselines*

The mention rule fires when the source label occurs in the target card. TF-IDF cosine uses word and bigram features of the same card text. A linear SVM and a logistic regression are fit on the training pairs of split E. The stack used in the tables is a separate logistic regression on three features of each directed pair: the mention indicator, verbalized confidence from the labels prompt, and verbalized confidence from the card-plus-path prompt. Its coefficients are fit on the training pairs, and its threshold is chosen on validation alone, on a grid of width 0.05. A mention-graph analogue of RefD, whose related set is the skills mentioned in a card, reaches PR-AUC 0.141. This construction is a graph-level reformulation of the mention rule,

and the one-line rule remains stronger. The shared-path orientation of the taxonomy prefix ranks below chance (AUC 0.163) because most skills sit at path length two and gold edges often cross the top-level branches. Reversing that orientation ranks above chance; precision–recall stays at the base rate because of ties.

### 5.3. *Author-labeled sample*

Two annotators and one adjudicator, the three authors, labeled 240 pairs on a four-way type and a direction. They saw labels, descriptions, taxonomy paths, and retrieved spans, and they did not see gold or model scores. The sheet mixed an 80-pair overlap with pairs unique to each annotator (seed 42). On the overlap, raw agreement is 0.875 and Cohen's $\kappa$ is 0.79. The adjudicated labels contain 58 direct pairs, 151 none, 31 ambiguous, and no transitive pair. Binary metrics treat direct as positive. A second sheet oversampled gold shortcuts and edges with an alternate path. Agreement there is 0.812 ($\kappa = 0.72$), and the transitive label was used once. The type field is therefore a schema slot rather than a measured classifier. Selective-review accuracy uses the first sheet only.

## 6. Results

### 6.1. *RQ1: the card as evidence*

Presenting the skill card as evidence improves ranking for `gpt-4o-mini` and lowers direction accuracy (Table 4). ROC-AUC rises from 0.834 to 0.886 and PR-AUC from 0.085 to 0.099. Adding the taxonomy path leaves both ranking metrics at the level of the card prompt. F1 at the fixed cut 0.5 falls because the card makes the model attach edges densely (recall 0.98 at precision 0.041). Under a 2% base rate that cut is a diagnostic. The validation-chosen threshold 0.55, applied to scaled confidence, recovers F1 0.190 for the card prompt and 0.203 for the card-plus-path prompt. A paired bootstrap of the card prompt against labels gives $\Delta$ROC-AUC 0.052 (0.043–0.062) and $\Delta$PR-AUC 0.014 (0.009–0.020), positive in all 1,000 resamples (Table 5). The AUC and PR-AUC intervals of the two evidence prompts overlap, so the taxonomy path adds no further ranking gain. Direction accuracy moves the other way: 0.761 for labels, 0.667 for cards, and 0.715 once the path is restored.

A shuffled-card control separates the heading from the content. On a gold-complete sample of the 1,003 test positives and 4,012 random negatives, the matched card scores ROC-AUC 0.887. A card taken from another pair scores 0.685, below the labels-only prompt on that sample (0.836). The gain tracks the matched card. The same format lowers ranking for Qwen 2.5 7B (Section 6.5).

### 6.2. *RQ2: mention and the stack*

The mention rule reaches PR-AUC 0.151, above every standalone model row in Table 4. TF-IDF is a strong ROC ranker and a weak precision–recall ranker. Neither

Table 4. Pair ranking on split E test (49,608 pairs, 1,003 positives), `gpt-4o-mini`. AUC and PR-AUC rank verbalized confidence. F1 at $\tau$ for the two evidence prompts uses the scaled value. The stack row is fit on the training pairs; $\tau$ is chosen on validation.

| Condition | F1@0.5 | F1@$\tau$ | $\tau$ | AUC | PR-AUC | DA |
|---|---|---|---|---|---|---|
| Mention | 0.339 | 0.339 | 0.50 | 0.620 | 0.151 | — |
| TF-IDF | 0.044 | — | — | 0.808 | 0.070 | — |
| Jaccard | 0.005 | — | — | 0.564 | 0.027 | — |
| Labels | 0.163 | 0.163 | 0.50 | 0.834 | 0.085 | **0.761** |
| Cards | 0.079 | 0.190 | 0.55 | **0.886** | 0.099 | 0.667 |
| Cards+path | 0.071 | 0.203 | 0.55 | 0.880 | 0.101 | 0.715 |
| Stack | — | **0.336** | 0.20 | 0.898 | **0.260** | — |
| Linear SVM | 0.014 | 0.111 | 0.30 | 0.697 | 0.073 | — |

Table 5. Bootstrap 95% intervals on split E test (1,000 resamples, seed 42) for the verbalized-confidence scores in Table 4.

| Condition | AUC | PR-AUC | F1@$\tau$ |
|---|---|---|---|
| Mention | [0.608, 0.634] | [0.130, 0.175] | [0.309, 0.372] |
| Labels | [0.820, 0.846] | [0.077, 0.093] | [0.152, 0.175] |
| Cards | [0.878, 0.893] | [0.092, 0.108] | [0.179, 0.202] |
| Cards+path | [0.872, 0.888] | [0.093, 0.111] | [0.190, 0.217] |

the linear SVM nor class-weighted logistic regression passes the mention rule. A simple disjunction of mention and the taxonomy prompt at $\tau = 0.55$ improves recall (0.616) at the cost of precision (0.126). The combination reported below is the train-fit stack defined in Section 5. On the test split it reaches ROC-AUC 0.898 and PR-AUC 0.260. At its validation threshold ($\tau = 0.20$) precision is 0.523, recall is 0.247, and F1 is 0.336. Mention F1 at 0.5 remains slightly higher (0.339), so the stack wins on ranking rather than at every operating cut. The model scores become useful for ranking once they are combined with the lexical rule. Pairs whose labels overlap lexically are scored highly even without evidence; Appendix D shows four illustrative pairs.

### 6.3. *RQ3: two graphs*

Structural constraints turn the scaled taxonomy-prompt scores into one acyclic graph, and the same constraints turn the train-fit stack into another (Table 6). The taxonomy graph keeps 8,896 edges at precision 0.102, full-gold recall 0.157, and graph F1 0.124. Before cycle breaking, the same scores keep 31,670 directed edges and recover more gold at lower precision. Cycle breaking guarantees acyclicity at a substantial cost in recall: test recall falls from 0.886 to 0.344 (345 of 1,003 gold test edges). Direction accuracy on those 345 edges is 0.554, below the score-level figure of 0.715 on all gold test edges, because the deleted edges are not a random subset.

The closure of the taxonomy graph reaches 1,220 gold edges (0.210 of unique gold). Appendix B records how many accepted edges also have an alternate path.

The stack graph is the higher-precision output. It keeps 1,334 edges at precision 0.530, full-gold recall 0.122, and graph F1 0.198. Recall on the original 1,003 test gold edges is 0.245 (246 edges), below the taxonomy graph. Precision is about five times higher, and graph F1 is higher. An audit that prefers fewer false edges would take the stack graph; an audit that prefers coverage would take the taxonomy graph. On the stack, both the acceptance cut and the cycle order use the logistic probability. The primary pool still bounds every graph built only from it: coverage is 48.3%, and the taxonomy graph reaches 0.325 of the gold edges inside the pool because cycle breaking deletes true positives. Appendix E reports the same constraints on a wider candidate pool and on a graph that also includes the training pairs.

Table 6. Directed graphs on 5,797 unique gold edges. F1 is full-gold graph F1. The taxonomy rows threshold scaled $\tilde{p}_{AB}$ at 0.55. The stack row thresholds the train-fit logistic probability at 0.20. R test uses the 1,003 gold test edges.

| Graph | $\|E\|$ | Prec. | F1 | R full | R pool | R val+test | R test |
|---|---|---|---|---|---|---|---|
| Cards+path at $\tau$ | 31,670 | 0.077 | 0.131 | 0.423 | 0.875 | 0.884 | 0.886 |
| Taxonomy graph | 8,896 | 0.102 | 0.124 | 0.157 | 0.325 | 0.344 | 0.344 |
| Stack graph | 1,334 | 0.530 | 0.198 | 0.122 | 0.253 | 0.238 | 0.245 |

The threshold $\tau = 0.55$ was fixed by maximizing pair-level F1 of scaled confidence on the validation split, before a graph F1 was computed. Table 7 then rebuilds a graph from the validation pairs alone (192 gold edges) across nearby thresholds. At $\tau = 0.60$, graph F1 rises from 0.148 to 0.242 because precision rises from 0.081 to 0.165 while recall falls from 0.813 to 0.453, and the graph shrinks from 1,921 edges to 528. Pair F1 and this graph F1 select different operating points, and 192 gold edges are too few to replace the pair-level cut that defines the published graphs. Table 7 makes this trade-off explicit. The margin $\mu \in \{0.6, 0.8, 1.0\}$ leaves every row unchanged: when $\mu \leq 1$ and $\tau \geq 0.5$, the weaker direction is always pushed below $\tau$. The 0.05 band and the cross-branch flag mark pairs and do not delete edges.

Table 7. Validation-only graphs from scaled Cards+path scores (8,611 pairs, 192 gold edges, $\mu = 0.8$). The same edge counts appear at $\mu \in \{0.6, 1.0\}$. These graphs are a sensitivity display, separate from Table 6.

| $\tau$ | $\|E\|$ | Prec. | Recall | F1 |
|---|---|---|---|---|
| 0.50 | 2,623 | 0.064 | 0.875 | 0.119 |
| 0.55 | 1,921 | 0.081 | 0.813 | 0.148 |
| 0.60 | 528 | 0.165 | 0.453 | 0.242 |
| 0.70 | 21 | 0.048 | 0.005 | 0.009 |

### 6.4. *Calibration*

Temperature scaling barely moves expected calibration error on the taxonomy prompt, from 0.306 to 0.299 (Table 8). The map divides the logit by a temperature and has no intercept, so it cannot remove a systematic over-positive bias at a 2% base rate. Platt scaling and a three-parameter beta map drive ECE to about 0.001 by sending almost every score to that base rate. Because those maps are monotone, a threshold rechosen on validation would leave pair F1 and AUC unchanged. They are poor graph-stage maps: Algorithm 1 compares scores with absolute constants, and a map that collapses to the base rate makes those constants meaningless. The JSON values are discrete in steps of 0.1, so many pairs share a rank. That granularity affects calibration error and the order of cycle deletion.

Table 8. Calibration on split E test. Temperature is fit on validation. ECE and Brier use verbalized confidence, raw and scaled.

| Condition | $T$ | ECE raw | ECE scaled | Brier raw / scaled |
|---|---|---|---|---|
| Labels | 3.2 | **0.086** | 0.109 | 0.055 / 0.064 |
| Cards | 3.9 | 0.282 | 0.273 | 0.174 / 0.146 |
| Cards+path | 3.3 | 0.306 | 0.299 | 0.178 / 0.160 |

### 6.5. *A second model family*

The card effect reverses on Qwen 2.5 7B (Table 9). On the test split, ROC-AUC falls from 0.803 for labels to 0.654 for cards; the taxonomy path recovers part of the drop (0.696) and remains below labels. PR-AUC follows the same order, while direction accuracy rises. Validation shows the same pattern. Whether a taxonomy card helps ranking therefore depends on the model. The comparison confounds model family with model size: one system is a proprietary model and the other is an open 7B model. A separate Qwen3 27B pass on the validation pairs, unused in any graph, ranks by the directed probability at AUC 0.711 and by verbalized confidence at AUC 0.202. An inverted confidence field would change which edges cycle breaking removes, which is why that pass stays out of Algorithm 1. On `gpt-4o-mini` the same confidence field has AUC 0.880 (Appendix C).

### 6.6. *Author labels*

On the 240 author labels, uncertainty sampling at a ten-percent budget (24 pairs) reaches accuracy 0.808, against 0.796 for a random slice of the same size. The bootstrap interval on that gap includes zero. At twenty and thirty percent the gaps are +0.054 (0.017–0.092) and +0.067 (0.021–0.096). Those intervals exclude zero, so on this author-labeled sample uncertainty sampling shows a small positive signal. The comparison was specified after the labels existed, and the annotators had seen

Table 9. Qwen 2.5 7B. F1 is at 0.5. AUC and PR-AUC rank verbalized confidence. Direction accuracy compares the two directed probabilities.

| Condition | Split | F1@0.5 | AUC | PR-AUC | DA |
|---|---|---|---|---|---|
| Labels | validation | 0.067 | **0.814** | **0.080** | 0.839 |
| Cards | validation | 0.045 | 0.652 | 0.034 | 0.901 |
| Cards+path | validation | 0.044 | 0.710 | 0.042 | **0.943** |
| Labels | test | **0.061** | **0.803** | **0.062** | 0.809 |
| Cards | test | 0.041 | 0.654 | 0.032 | 0.908 |
| Cards+path | test | 0.040 | 0.696 | 0.036 | **0.947** |

the retrieved spans. Table 10 compares the same scores with the 240 labels. The slice is denser in positives than the ontology, and it agrees with the ranking pattern of `gpt-4o-mini` on split E.

Table 10. The 240 author-labeled pairs. Positive means direct; none and ambiguous count as negative.

| Condition | F1@0.5 | F1@0.55 | AUC | PR-AUC |
|---|---|---|---|---|
| `gpt-4o-mini` Labels | **0.714** | **0.768** | 0.872 | 0.634 |
| `gpt-4o-mini` Cards | 0.435 | 0.641 | 0.899 | 0.612 |
| `gpt-4o-mini` Cards+path | 0.494 | 0.699 | **0.949** | **0.748** |
| Qwen 2.5 7B Cards | 0.395 | 0.459 | 0.720 | 0.550 |

## 7. Discussion

The card effect splits by model. For `gpt-4o-mini`, restating the skill's own card under an evidence heading improves ranking, and a card drawn from another pair removes that gain. For Qwen 2.5 7B the same restatement lowers ranking on both validation and test. Work on verbalized confidence finds that the number a model writes down is sensitive to the prompt and is often over-confident [25,26]. A reading consistent with that evidence is that the two models use a prompted evidence block differently: one treats the matched card as support, and the other lets the repeated text inflate both directions. Because only the verbalized outputs were recorded, instruction-following effects cannot be separated from context-length effects. Direction accuracy and ranking move in opposite directions for both models, so a prompt that helps one of them can hurt the other.

On this release the lexical rule is the strong prior. Gold edges often pair a knowledge label with an action that repeats it, which is why mention reaches a higher standalone PR-AUC than any single prompt. The language model still recovers gold edges the rule misses, and the stack is the way those two signals combine. LECDPR [1] invests the retrieved text in a graph neural network over concept–

document interactions. ProPRL [2] trains against bidirectional over-confidence; Algorithm 1 applies a related penalty only at inference, after the scores exist. Le and Abel [5] generate prerequisite names and align them. The tool here scores directed pairs inside a fixed candidate pool and returns an edge with a span. These systems address different tasks, and their F1 tables answer different questions.

For a practitioner the choice is between the two graphs in Table 6. The stack graph is the output to hand to an audit that prefers fewer false edges. The taxonomy graph is the output when missing a gold edge is more costly than an extra edge, with the understanding that cycle breaking has already removed many true positives. Both graphs are bounded by candidate generation. A pool that also kept single-token label overlap would cover 74.9% of gold edges, at the cost of a substantially larger candidate set (about 562k pairs).

The result is tied to ESCO cards under an unseen-concept split. Extending either model to lecture slides, Wikipedia lead sections, or a taxonomy whose gold edges are less lexical requires a second corpus. External raters who have not seen the retrieved spans would be needed before the human sample could support a selective-review policy.

## 8. Limitations

The primary pool contains fewer than half of the gold edges, so a perfect classifier on that pool would still miss most of the ontology. Wikipedia sitelinks and lecture slides are too sparse, or too incomplete in the public files, to serve as a second corpus. The human sample was labeled by the authors after they had seen retrieved spans, and the transitive type was almost never chosen, so the type field remains unchecked. A slice of skills introduced after the model cutoff is not available: the identifier difference against ESCO v1.1.1 contains no gold positive in the test split. Label permutation drives AUC to chance (Appendix C), which shows that the metrics respond to label structure. The two scored models differ in both family and size, so the card effect cannot yet be attributed to family alone. The two graphs use thresholds chosen on validation for pair F1. Table 7 shows that a different objective, graph F1 on 192 validation gold edges, prefers a sparser cut.

## 9. Conclusion

Restating a skill card as evidence improves pairwise ranking for `gpt-4o-mini` and lowers it for Qwen 2.5 7B, and it lowers direction accuracy for the first model. A mention rule is the stronger standalone ranker; stacking it with the model scores is what makes those scores useful. Temperature scaling leaves calibration almost unchanged and is kept because the graph constraints use absolute cuts. The resulting taxonomy graph recovers more gold edges, and the stack graph is more precise. The next useful measurement is a second corpus whose documents are not the skill cards themselves, scored with the same metric-to-field map.

### Data and Code Availability

ESCO-PrereqSkill is public [5]. A reproduction pack contains the conversion and scoring scripts, the prompts, the configuration (seed 42, unseen-concept fraction 0.2), the held-out skill identifiers, the two 240-pair human sheets, and the pair scores needed to recompute the test tables. The pack is public at `https://github.com/tuanymc/p2a-prerequisite-graphs`. The full prompt-and-span logs will be posted with the camera-ready version.

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The authors declare that they have no conflict of interest.

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### Appendix A. Scoring Prompts

The system message is shared by the three prompts:

```
You judge directed prerequisite relations.
Return JSON only. Do not invent quotations.
Cite only provided evidence_ids.
Fields:
- p_A_to_B, p_B_to_A: numeric floats in [0,1],
  never strings or labels
- relation_type: direct | transitive | none | ambiguous
- direction: A_to_B | B_to_A | none
- confidence_raw: numeric float in [0,1]
- rationale: short string
- evidence_ids: array of provided ids
```

The user message begins with the two skill records. Cards+path then inserts `Taxonomy A:` and `Taxonomy B:`. Cards and Cards+path append retrieved windows as `Evidence A→B:` and `Evidence B→A:`. Labels replaces those blocks with "No retrieved evidence. Do not fabricate spans." In the released logs, Labels, Cards, and Cards+path are the conditions B1, B3, and B5. The directed probabilities are $\hat{p}_{AB}$ and $\hat{p}_{BA}$. The verbalized confidence is $\hat{c}_{AB}$.

## Appendix B. Reproduction Notes

On the reported full-pool run, reverse-conflict suppression flags 10,699 of the 31,670 edges that pass $\tau = 0.55$. Cycle breaking then deletes 22,774 edges and leaves the 8,896-edge graph. The batched pass deletes 21,375 edges; a sequential cleanup deletes the remaining 1,399 over 153 rounds. When several edges in one component share a confidence, the deleted edge is the one with the smaller source identifier, then the smaller target identifier. A cross-branch flag marks 178 high-confidence pairs and deletes none. Every scored pair has a card window, so the missing-evidence flag is never raised. Of 7,985 accepted edges that miss a direct gold link, four lie on the gold transitive closure. Of the accepted edges, 16.7% have an alternate path, and 173 of those paths are gold-direct. A transitive reduction of the gold graph finds 38 shortcuts (0.7% of 5,797). The second author sheet assigned `transitive` once, so that type stays unevaluated as a classifier.

## Appendix C. Score Fields and Robustness

On the 49,608 test pairs, verbalized confidence for the taxonomy prompt has ROC-AUC 0.880 and PR-AUC 0.101, which are the Cards+path entries in Table 4. The directed probability on the same rows has ROC-AUC 0.894 and PR-AUC 0.120. The rank correlation of the two fields is 0.880. Thresholding the scaled directed probability at 0.55 gives pair F1 0.143 (precision 0.078, recall 0.887). That cut is the pair-level form of edge acceptance. Labels and Cards show the same pattern: confidence matches Table 4 (AUC 0.834 and 0.886), while the directed probability has AUC 0.859 and 0.885. Cycle deletion on the taxonomy graph follows scaled confidence. Replacing that order with the directed probability would define a different graph; Appendix B describes the reported graph.

Labels F1 is 0.163 on split E and 0.156 on the closed split A. A zero-shot scorer has no weights for the split to change. Permuting the test labels (20 shuffles, seed 42) drops ROC-AUC to 0.50 (interval 0.49–0.51) and PR-AUC to the 2% base rate for labels, cards, the taxonomy prompt, and mention. This permutation test confirms that the metrics are sensitive to label structure; it does not, however, establish whether the model memorized ESCO. ESCO v1.2 (May 2024) is after the October 2023 knowledge cutoff of `gpt-4o-mini`. The identifier difference against the September 2022 public list yields one new skill, ten test pairs, and zero gold positives.

## Appendix D. Illustrative Pairs

These examples are illustrative and were not sampled to be representative. Values are $\hat{p}_{AB}/\hat{p}_{BA}$. *Soldering techniques* before *apply soldering techniques* is 0.8/0.8 under Labels and only slightly higher once the card is shown, with both directions high. *Types of concrete pumps* before *install concrete pumps* moves from 0.0/0.0 to 0.7/0.5. The same spans raise a same-verb distractor, *assemble mechatronic units*

before *assemble hardware components*, from 0.3/0.2 to 0.7/0.5, and they raise both directions on a gold-reverse pair, *design user interface* before *use methodologies for user-centered design*. An automatic audit of gold edges absent from the primary pool finds no remaining edge whose one label is a token-subset of the other, 22 missed edges that share a bigram, and 1,548 missed edges with at least one shared token.

## Appendix E. Candidate-Pool Expansion

TF-IDF top-16 neighbors of the card text cover 3,636 unique gold edges (0.627). The union with the primary pool covers 3,956 (0.682). The 63,718 new pairs were scored once with the taxonomy prompt (`gpt-4o-mini`, temperature 0, zero errors). Each pool then passes separately through Algorithm 1. Table 11 reports the three acyclic graphs. We expected full-gold recall to rise by less than 0.02, or the interval to include zero. Both pools rise by about 0.03, and the intervals exclude zero. On the original 1,003 test gold edges the primary graph keeps 345, the neighborhood graph keeps 300, and the union keeps 309. Because the neighborhood omits some primary-pool pairs, the pre-cycle edge set already falls from 889 to 802 true positives, and cycle breaking leaves 300. The union keeps all 889, then cycle breaking removes 36 edges the primary graph had kept. Of the 172 additional full-gold true positives in the neighborhood graph, 56 lie in an expanded test set of 1,425 edges and 116 lie in training or validation gold. The primary pool contains none of the 422 newly added test gold edges, so those edges count against it before the classifier runs. A top-32 neighborhood was left unscored; top-16 already shows the recall gain and the cost on the original test gold. Including the training pairs in the taxonomy graph raises full-gold recall relative to a graph built only on validation and test, and cycle breaking on that denser graph lowers test recall from 0.423 to 0.344.

Table 11. Full-gold recovery after scoring the TF-IDF top-16 pairs absent from the primary pool (5,797 unique gold). R original is recall on the 1,003 gold test edges already inside the primary pool. Intervals are paired bootstrap percentiles, 1,000 resamples, seed 42.

| Pool | $\|E\|$ | Prec. | F1 | R full | R original | ΔR | 95% interval |
|---|---|---|---|---|---|---|---|
| Primary | 8,896 | 0.102 | 0.124 | 0.157 | 0.344 | 0 | — |
| TF-IDF-16 | 9,633 | 0.112 | 0.140 | 0.187 | 0.299 | +0.030 | 0.022–0.037 |
| Union | 10,325 | 0.104 | 0.133 | 0.185 | 0.308 | +0.028 | 0.021–0.035 |

## Appendix F. Recovered Slices

On 4,640 pairs recovered by single-token overlap outside the primary pool (1,544 gold, positive rate 0.333), repeating the card raises `gpt-4o-mini` ROC-AUC by 0.127 (0.115–0.138) and PR-AUC by 0.110 (0.093–0.127). On Qwen 2.5 7B the same

change lowers ROC-AUC by 0.058 (0.048–0.067) and PR-AUC by 0.117 (0.099–0.132). The interaction, the Qwen change minus the `gpt-4o-mini` change, is −0.184 on ROC-AUC (−0.200 to −0.169) and −0.226 on PR-AUC (−0.251 to −0.205). These slices are denser than Table 4. Inside the TF-IDF top-16 neighbors (3,468 pairs, 1,156 gold, the same positive rate) the card raises `gpt-4o-mini` ROC-AUC by 0.048 (0.035–0.060) and leaves PR-AUC unchanged (−0.001; the interval includes zero), while it lowers Qwen on both (ROC −0.070, PR −0.078). The top-32 neighbor slice has the same signs.

## Appendix G. Token Cost

A 400-pair sample of taxonomy-prompt inputs for `gpt-4o-mini` averaged 401 input tokens (90th percentile 448). We budget 160 output tokens. At the October 2026 list price of $0.15 / $0.60 per million input and output tokens, the validation-plus-test pass (58,219 pairs) is about $9, and the additional 77,507 training pairs are about $12. The TF-IDF expansion (63,718 pairs) is about $10. Labels on the training pairs and the new pairs (141,225 calls, zero errors) is about $22 on the same budget. Qwen token bills were not retained. Wall-clock time for the parallel score loop was about 3.8 h (labels), 5 h (cards), 4.8 h (taxonomy prompt, validation and test), and 6.9 h (taxonomy prompt, training).

## References

1. K. Xiao, L. Zheng, X. He, M. Zhang, Z. Huang and Y. Zhang, LECDPR: LLM enhancement and concept-document interactive modeling for prerequisite relation prediction, in *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence IJCAI-2026*, (International Joint Conferences on Artificial Intelligence Organization, September 2026), pp. 5092–5100.
2. X. Cheng, J. Wang, C. He, R. Dong and Q. Guan, ProPRL: Property-aware prerequisite relation learning in educational knowledge graphs, arXiv:2608.03006, (2026).
3. X. Li, F. Min, M. Li, C. Zhang and Z. Wang, AutoPRE: Discovering concept prerequisites with LLM agents, in *Natural Language Processing and Chinese Computing* (Springer Nature Singapore, November 2025), pp. 199–211.
4. X. Cheng, Z. Zhang, J. Wang, L. Fang, C. He, Q. Guan, S. Pan and W. Luo, GraphRAG-induced dual knowledge structure graphs for personalized learning path recommendation, *Proceedings of the AAAI Conference on Artificial Intelligence* **40** (March 2026) 14610–14620.
5. N. L. Le and M.-H. Abel, How well do LLMs predict prerequisite skills? Zero-shot comparison to expert-defined concepts, in *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (IEEE, October 2025), pp. 7245–7250.
6. C. Liang, Z. Wu, W. Huang and C. L. Giles, Measuring prerequisite relations among concepts, in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2015), pp. 1668–1674.
7. C. Liang, J. Ye, Z. Wu, B. Pursel and C. Giles, Recovering concept prerequisite relations from university course dependencies, *Proceedings of the AAAI Conference on Artificial Intelligence* **31** (February 2017).

8. Y. Yang, H. Liu, J. Carbonell and W. Ma, Concept graph learning from educational data, in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining WSDM 2015*, (ACM, February 2015), pp. 159–168.
9. H. Liu, W. Ma, Y. Yang and J. Carbonell, Learning concept graphs from online educational data, *Journal of Artificial Intelligence Research* **55** (April 2016) 1059–1090.
10. L. Pan, C. Li, J. Li and J. Tang, Prerequisite relation learning for concepts in MOOCs, in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Association for Computational Linguistics, 2017), pp. 1447–1456.
11. S. Roy, M. Madhyastha, S. Lawrence and V. Rajan, Inferring concept prerequisite relations from online educational resources, *Proceedings of the AAAI Conference on Artificial Intelligence* **33** (July 2019) 9589–9594.
12. K. Xiao, Y. Bai and Z. Wang, Extracting prerequisite relations among concepts from the course descriptions, *International Journal of Software Engineering and Knowledge Engineering* **32** (April 2022) 503–523.
13. J. Sun, Y. He, Y. Xu, J. Sun and G. Sun, A learning-path based supervised method for concept prerequisite relations extraction in educational data, in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management CIKM '24*, (ACM, October 2024), pp. 2168–2177.
14. Y. Zhang, X. Qu, S. Liu, Y. Pang and X. Shang, Multiscale Weisfeiler-Leman directed graph neural networks for prerequisite-link prediction, *IEEE Transactions on Knowledge and Data Engineering* **37** (June 2025) 3556–3569.
15. D. Mazumder, J. H. Paik and A. Basu, A graph neural network model for concept prerequisite relation extraction, in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management CIKM '23*, (ACM, October 2023), pp. 1787–1796.
16. J. Xia, M. Li, Y. Tang and S. Yang, Course map learning with graph convolutional network based on AuCM, *World Wide Web* **26** (July 2023) 3483–3502.
17. F. Chiarello, G. Fantoni, T. Hogarth, V. Giordano, L. Baltina and I. Spada, Towards ESCO 4.0 – is the European classification of skills in line with Industry 4.0? A text mining approach, *Technological Forecasting and Social Change* **173** (December 2021) p. 121177.
18. M. Hosseinioun, F. Neffke, L. Zhang and H. Youn, Skill dependencies uncover nested human capital, *Nature Human Behaviour* **9** (February 2025) 673–687.
19. Y. Dai, M. Yoshikawa and K. Sugiyama, Prerequisite-aware course ordering towards getting relevant job opportunities, *Expert Systems with Applications* **183** (November 2021) p. 115233.
20. P. Chen, Y. Lu, V. W. Zheng, X. Chen and B. Yang, KnowEdu: A system to construct knowledge graph for education, *IEEE Access* **6** (2018) 31553–31563.
21. N. L. Le, M.-H. Abel and B. Laforge, From learning resources to competencies: LLM-based tagging with evidence and graph constraints, in *Artificial Intelligence in Education* (Springer Nature Switzerland, June 2026), pp. 616–631.
22. J. Shen, Z. Shen, C. Xiong, C. Wang, K. Wang and J. Han, TaxoExpan: Self-supervised taxonomy expansion with position-enhanced graph neural network, in *Proceedings of The Web Conference 2020* (Association for Computing Machinery, Taipei, Taiwan, 2020), pp. 486–497.
23. I. Li, A. R. Fabbri, R. R. Tung and D. R. Radev, What should I learn first: Introducing LectureBank for NLP education and prerequisite chain learning, *Proceedings of the AAAI Conference on Artificial Intelligence* **33** (July 2019) 6674–6681.

24. C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, On calibration of modern neural networks, in *Proceedings of the 34th International Conference on Machine Learning Proceedings of Machine Learning Research* **70**, (PMLR, 2017), pp. 1321–1330.
25. K. Tian, E. Mitchell, A. Zhou, A. Sharma, R. Rafailov, H. Yao, C. Finn and C. D. Manning, Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback, in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, Singapore, 2023), pp. 5433–5442.
26. M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He and B. Hooi, Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs, in *International Conference on Learning Representations* (OpenReview, 2024).
27. P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel and D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks, in *Advances in Neural Information Processing Systems* **33**, (Curran Associates, Inc., 2020).
28. Y. Zhu and H. Zamani, Predicting prerequisite relations for unseen concepts, in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2022), pp. 8542–8548.
29. C. Alzetta, I. Torre and F. Koceva, Annotation protocol for textbook enrichment with prerequisite knowledge graph, *Technology, Knowledge and Learning* **29** (March 2024) 197–228, Published online 21 September 2023.